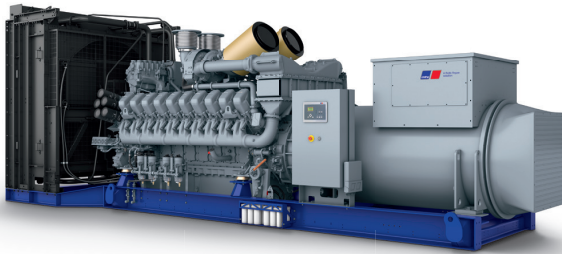




Diesel Generator Set

mtu 20V4000 DS4000

400 V - 11 kV/50 Hz/data center continuous power/
NEA (ORDE) + Tier 2 optimized/20V4000G44LF/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System rating: 3600 kVA - 3630 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 100% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Electrical driven radiators
- Medium voltage alternators
- Low voltage alternator

Emissions

- Tier 2 optimized engine
- NEA (ORDE) optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer	mtu		Total oil system capacity: l	390
Model	20V4000G44LF		Engine jacket water capacity: l	260
Type	4-cycle		Intercooler coolant capacity: l	50
Arrangement	20V			
Displacement: l	95.4		Combustion air requirements	
Bore: mm	170		Combustion air volume: m³/s	4.4
Stroke: mm	210		Max. air intake restriction: mbar	30
Compression ratio	16.4			
Rated speed: rpm	1500		Cooling/radiator system	
Engine governor	ADEC (ECU 9)		Coolant flow rate (HT circuit): m³/hr	80
Max power: kWm	3007		Coolant flow rate (LT circuit): m³/hr	50
Air cleaner	dry		Heat rejection to coolant: kW	1045
			Heat radiated to charge air cooling: kW	835
			Heat radiated to ambient: kW	105
			Fan power for electr. radiator (40°C): kW	105
			Exhaust system	
			Exhaust gas temp. (after engine, max.): °C	440
			Exhaust gas temp. (before turbocharger): °C	620
			Exhaust gas volume: m³/s	10.6
			Maximum allowable back pressure: mbar	50
			Minimum allowable back pressure: mbar	-

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) + Tier 2 optimized		
		without radiator		
		kWel	kVA*	AMPS
Leroy Somer LSA54.2 ZL17 (LV Leroy Somer standard)	400 V	2880	3600	5196
Leroy Somer LSA54.2 ZL12 (Med. volt. Leroy Somer)	11 kV	2904	3630	191
Marathon 1040FDH7105 (Medium volt. marathon)	11 kV	2904	3630	191
Leroy Somer LSA54.2 ZL14 (MV Leroy Somer oversized)	11 kV	2904	3630	191

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power and are approximate values.